JEE Organic Chemistry Reaction Mechanisms & Reagent Roadmaps

Target Exam: JEE Main & JEE Advanced 2026 | Stream: Engineering | Subject: Chemistry

Verified Official Academic Preparation Material - TheCampusNova Learning Portal

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Key Topics Covered: Named Reactions, GOC, Aldehydes, Ketones & Amines

1. General Organic Chemistry: Electronegativity, Inductive Effect (+I / -I orders), Hyperconjugation & Resonance stability
2. Aldol Condensation: Aldehydes/ketones with alpha-H in dilute NaOH yield beta-hydroxy carbonyls -> alpha,beta-unsaturated
3. Cannizzaro Reaction: Non-enolizable aldehydes (e.g. Formaldehyde, Benzaldehyde) in 50% KOH undergo disproportionation
4. Reagent Hacks: $LiAlH_4$ (strong, reduces esters/acids/carbonyls to alcohols) vs $NaBH_4$ (mild, reduces only aldehydes/ketones)
5. Reagent Hacks: PCC / PDC oxidizes primary alcohols strictly to aldehydes without over-oxidation to carboxylic acids
6. Exam Strategy: Master NCERT reagent charts; 90% of Organic Chemistry questions in JEE Main are direct named transformations.